

Chaokang Jiang

Homepage · Scholar

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1998.01

World-Model Algorithm



Education

2020.09 ~ 2023.06 M.Sc.-Control Sci.&Eng.-SJTU/CUMT Adv.: H. Wang (Dean, Pujiang Int'l), Y. Miao IRMV Lab

Work Experience

2023.06 ~ 2025.03 Perception Algorithm Phigent Robotics Beijing
2025.03 ~ Present World-Model Algorithm (top group perf.) Bosch (XC-CN) Shanghai

Project Experience

- **Generative Closed-Loop AD Simulation** Bosch (XC-CN) · 2026.01 ~ Present
 - 1) Led 7V surround-video generation end-to-end (Map→WorldSim→7V); distillation gave **13.9×** faster inference;
 - 2) built a 3-stage OmniDream scheme: ① bidirectional teacher (global temporal) → ② causal student (autoregressive) → ③ distillation + self-forcing student (streaming) — industry-first real-time streaming 7V video generation;
 - 3) designed a SmartAgent × vector-world-model framework for semantically-controllable, closed-loop AD simulation.
- **Two-/One-Stage Perception Iteration** Bosch (XC-CN) · 2025.03 ~ 2025.12
 - 1) Built static OneModel (features 2→8: laneboundary / roadedge → roadmarker / centerline / goalpoint / GRE / freespace / drivepath); optimized Training/Inference/Evaluation frameworks; shipped to **3** production Platform Models over 20+ major releases;
 - 2) delivered a DYO & Static two-in-one OneModel (one-stage); **3**) led dynamic-model quantization & Mono3D releases.
- **Controllable Surround Video Generation** Phigent Robotics · 2024.07 ~ 2025.03
 - 1) Built a controllable generation/editing pipeline for internal surround-view data (MagicDrive), supporting 4-fisheye/7V/11V;
 - 2) iterated a multimodal surround-video framework (OpenSora-STDiT × MagicDrive): text/layout control across views/resolutions/durations, stable **241+** frame generation; validated on cone-line/truck-replacement scenarios.
- **Perception/Data-Gen Optimization & Sparse E2E PoC** Phigent Robotics · 2024.02 ~ 2024.08
 - 1) AutoLabelling: iterated a BEVFusion dynamic auto-labeler for better VRU precision and long-range recall;
 - 2) Perception toolchain: optimized multi-sensor calibration QC and pure-LiDAR 3D-detection efficiency;
 - 3) built a Pillar 3D velocity-flow model on J6E/Orin, with scene-flow auto-labeling and point-cloud temporal-prediction;
 - 4) supported 11V Camera × LiDAR Sparse BEV-Fusion design and maintained fused CUDA ops.
- **Stereo Road-Preview Magic-Carpet (DiSus System)** Phigent Robotics · 2023.06 ~ 2024.04
 - 1) Deployed road-segmentation models on TDA4, Horizon J5/J6E and NVIDIA Orin (float + QAT), balancing accuracy/efficiency into mass production; grew data 5W→**23W+** over 20+ releases; **2**) pre-researched a stereo depth-estimation model across architectures, pushing toward mass production.
- **More projects (see homepage, some content encrypted: 5896)**

Autonomous lawn-mower-robot perception (Positec), offline pure-vision 4D AutoLabeling system, etc.

Academic Achievement

- **10+** papers across ICML (Spotlight), NeurIPS, CVPR, ICCV, AAAI and T-PAMI, TII, TIM, TITS, AIS.
- **VectorWorld (ICML 2026 Spotlight)** Streaming vector world model for AD, dual SOTA in accuracy and efficiency.
- **3DSFLabelling (CVPR 2024)** Ultra-precise motion-flow labeling/augmentation, boosting perception to SOTA.
- **NeuroGauss4D-PCI (NeurIPS 2024)** 4D neural/Gaussian fields for point-cloud interpolation & prediction.
- **Pseudo-LiDAR Scene Flow (TII, Q1 Top)** First pure-vision scene-flow estimation via depth-consistency loss.
- **More:** DiffFlow3D(CVPR'24/T-PAMI), Mamba4D(CVPR'25), RegFormer(ICCV'23), TransLO(AAAI'23), Pseudo-LiDAR VO(TIM'23), SFGAN(AIS'22) — covering scene flow, odometry/registration, and 3D/4D perception and generation.

Expertise

PyTorch, mmdet, diffusers, transformers, CUDA, Docker, PCL; Vibe Coding; R&D-to-production; 10+ NSFC (General/Youth) grants.